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A Fail-Closed Point-Zero Audit of Open Data for UK Microtransaction Microsimulation: Reproducible Calibration Infrastructure and Non-Identification Results

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> **Immediate scope warning.** This is an exploratory, fail-closed computational audit. It does not produce estimates of microtransaction prevalence, real-world causality, regulatory effectiveness, clinical harm, national expenditure, or policy forecasts. It does not execute or compare policy scenarios.

Abstract

Microtransaction microsimulations can appear empirically grounded when they combine open aggregate sources, yet technical reproducibility does not establish behavioural identification or external validity. This study reports a deterministic, fail-closed point-zero audit of the UK population binding, evidence roles, assumptions, runtime consumption, and integrity gates implemented in the `microtx-sim` repository; repeatability is conditional on the audited local environment and exact hash-attested cache. The audited bundle, `uk-adults-2024-v1`, is explicitly `PARTIAL` and `campaign_ready=false`. The checkpoint initializes a fixed seed-101 synthetic cohort but executes no scenario, policy day, or simulation campaign. Of 12 repository-coded structural, plausibility, and identification gates, six pass and six fail; the overall status is `FAIL_CLOSED`. The framework attests declared bytes, cohort identity, UK selection, runtime binding, sex state, and initializer bounds, but authorizes no behavioural, causal, population, or regulatory claims. The central demographic diagnostic is a total-variation distance of TVD=0.033053, above the default engineering threshold of 0.020000. In addition, P(gamer | age, sex) is not identified, projected gamer/non-gamer labels are not behaviourally binding, and available play-time sources are incommensurable in construct, denominator, age scope, and sampling design. Making this empirical no-go decision visible and content-addressed is preferable to producing apparently credible but unidentified synthetic scenarios. ChatGPT GPT-5.6 Sol assisted code development, model design, and drafting; the author reviewed its outputs and accepts full responsibility.

Declaration of Generative AI Use. Generative AI tools were used in code development, model design, and the drafting, writing, and editing of this manuscript. ChatGPT GPT-5.6 Sol was used to support code development, model design, and manuscript drafting and writing. All AI-assisted code, model specifications, text, citations, and references were reviewed and verified by the author. The point-zero audit was executed computationally, and the reported results were not generated by AI.

Keywords: agent-based modelling; microtransactions; computational validation; reproducibility; calibration; open data; fail-closed design; data provenance; gaming regulation

1. Introduction

Microsimulation is attractive for studying video-game microtransactions because the mechanisms of interest operate at several levels at once. Individual play, purchase opportunities, liquidity, game design, firm conduct, and regulatory constraints may interact over time even when no single dataset observes the full system. A transparent computational model can make those mechanisms explicit and can expose where policy conclusions depend on assumptions. The same flexibility, however, creates a risk: a technically elaborate simulator may acquire an empirical appearance merely because it cites official statistics and other secondary datasets. Existing research treats microtransactions as a heterogeneous class and identifies varied monetisation mechanisms (Gibson et al., 2022; Petrovskaya & Zendle, 2022). The systematic-review evidence is largely cross-sectional and does not establish causal direction, which limits regulatory inference (Gibson et al., 2022).

The principal epistemic problem examined here is data composition without identification. Aggregate secondary sources can differ in population, denominator, period, construct, measurement mode, selection mechanism, and uncertainty. A population workbook may identify an age-by-sex margin but not gaming participation. A survey of gamers may describe self-reported play among those already classified as gamers but not $P(\text{gamer} \mid \text{age}, \text{sex})$ in the resident population. A selected trace-sharing cohort may contain rich within-person measures while remaining unsuitable for population transport. Runtime variables may also use different semantics from the observed measure: an initialized intention is not realised diary time. Combining these inputs without a declared measurement and transport model can therefore produce a simulation that is reproducible as software but not identified as an empirical representation of UK microtransaction behaviour. Related work on transportability formalizes the need to state assumptions when evidence is moved across populations; this paper does not perform causal transport analysis (Pearl & Bareinboim, 2014).

This study asks:

> **What can a content-addressed, fail-closed point-zero audit establish about the readiness of available secondary evidence to support a UK microtransaction microsimulation?**

The question is deliberately narrower than whether a microtransaction policy would be effective. Four claims are kept separate throughout:

1. **Computational reproducibility:** whether declared bytes, configuration, transformations, assignments, and outputs can be regenerated under the stated local conditions.
2. **Conformity to demographic margins:** whether the initialized, weighted synthetic cohort matches the declared ONS age-by-sex distribution within an engineering tolerance.
3. **Identification of behavioural parameters:** whether the evidence identifies quantities such as $P(\text{gamer} \mid \text{age}, \text{sex})$, purchase participation, spending, or harm, and whether compatible targets are consumed by outcome-changing runtime equations.
4. **External validity and policy-claim authorization:** whether any fitted model has been independently validated for its target population and is permitted to support population, causal, regulatory, clinical, expenditure, or forecasting claims.

Success at an earlier layer is necessary but not sufficient for a later one. Hash agreement can establish byte identity without establishing publisher authenticity or representativeness. Exact arithmetic can establish a synthetic allocation without reconstructing observed microdata. A plausible initialization can remain behaviourally unidentified. This hierarchy motivates a fail-closed design: when a required gate fails, the empirically labelled workflow stops and reports the failure instead of proceeding under an implicit assumption. Computational reproducibility supports scrutiny but does not itself establish simulation-model validity (Sandve et al., 2013; Sargent, 2013).

The repository contains a broader synthetic policy simulator, but that functionality is outside this paper's estimand. No campaign, policy scenario, policy day, causal contrast, sensitivity campaign, or empirical forecast is run or reported here. The object of study is the initialization boundary itself: a point-zero audit of what the partial calibration bundle, its lineage, and the runtime binding can and cannot establish.

2. Evidence Scope and Conceptual Boundaries

2.1 The partial evidence bundle

The machine-readable bundle `uk-adults-2024-v1` declares `status=PARTIAL` and `campaign_ready=false`. It contains 75 typed target rows, ten ONS age-by-sex population cells, and 15 content-addressed source records. These records include ONS population and time-use material, the DWP Family Resources Survey (FRS) (Department for Work and Pensions, 2025), Ofcom's *Online Nation 2024* report containing Ampere gaming figures, and Open Play v1.2.5. The bundle also contains proxy, unquantified, and normative records. Presence in the bundle does not itself make a record a runtime input, a fitting target, or an empirical validation criterion.

The schema separates estimand role from evidence status. Roles are operational rather than publisher-wide: a `CALIBRATION` target may enter a fit only when population, period, unit, conditioning set, transformation, and runtime construct are compatible; a `VALIDATION` target must remain outside fitting and requires a predeclared acceptance rule; and a `DIAGNOSTIC` target can reveal sign, order, scale, or incompatibility without identifying a population parameter. Evidence status is a separate field: `QUANTIFIED` records carry a value under their stated estimand, a `PROXY` stands for a related construct under an explicit assumption, `UNQUANTIFIED` records mark missing point targets, and `NORMATIVE` records mark value choices that cannot be estimated as if they were descriptive facts. Source-level roles are separate again and do not override target-level typing. The separation is consistent with longstanding accounts of empirical validation in agent-based modelling (Windrum, Fagiolo, & Moneta, 2007).

Table 1. Principal evidence families and epistemic status

Evidence family	Population and measure	Source role; target role; evidence status	Runtime consumption at point zero	Inference permitted here
ONS mid-2024 population	UK usual residents aged 18-64; counts by five age bands and source-recorded female/male categories	`Target role CALIBRATION; evidence QUANTIFIED`	Ten age-by-sex cells consumed by the point-zero sex binding; age is not resampled	Demographic margin and lineage only
ONS Time Use, March 2024	UK adults aged 18+; realised primary-activity minutes/day	Source role MIXED; gaming target role DIAGNOSTIC; evidence QUANTIFIED	Not connected to the life initializer	Construct and scale diagnostic only
ONS Time Use, Sep-Oct 2023	UK adults aged 18+; realised primary-activity minutes/day	`Target role encoded as VALIDATION; evidence QUANTIFIED; independent row-level holdout not verified`	Not consumed; no validation gate has passed	Encoded temporal contrast only; partition independence not verified

Evidence family	Population and measure	Source role; target role; evidence status	Runtime consumption at point zero	Inference permitted here
DWP FRS 2023/24	UK private households; rounded gross weekly household-income shares	Source role MIXED; income-share targets CALIBRATION; rounding-total targets DIAGNOSTIC; evidence QUANTIFIED	Not connected to individual monthly disposable income	Aggregate household margins only
Ofcom/Ampere Q2 2024	UK gamers aged 16-64; self-reported weekly play	`Target role encoded as VALIDATION; evidence QUANTIFIED; independent holdout not verified`	Not consumed	External diagnostic discrepancy; no validation gate passed
Open Play v1.2.5	Selected, qualified UK participants aged 18-40; self-report and platform traces	`Target role DIAGNOSTIC; evidence QUANTIFIED`	Excluded or not connected	Conditional sample description; no UK population transport
Gamer participation, transaction bridge, and harm parameters	Required behavioural or normative constructs	`Target role DIAGNOSTIC; evidence UNQUANTIFIED or NORMATIVE`	No empirical binding	No point estimate or policy claim

2.2 Populations and measures that must not be collapsed

The target demographic universe is UK usual residents aged 18-64 at mid-2024. It is not the same population as UK gamers aged 16-64 in the Ofcom/Ampere figure. It is also not the selected Open Play cohort, which covers qualified UK participants aged 18-40 who accepted an intensive trace-sharing protocol. These populations differ before any numerical comparison is made.

The measures differ as well. The ONS value of 16.8 minutes/day describes realised, unconditional primary-activity gaming among adults aged 18 and over (Office for National Statistics, 2024). The Ofcom report describes gamer-only weekly self-report attributed to Ampere Games-Consumer, Q2 2024 (Ofcom, 2024). Open Play's repository-derived weekly mean is an intake self-report among a selected subset of that study (Ballou et al., 2026). The simulator field is `intended_play_minutes`, an initialized intention. These quantities are not repeated measurements of one common estimand.

Open Play is therefore not described in this paper as a UK time-use diary. The published study design states that daily surveys and time-use diaries were US-only. The local cache nevertheless contains 43 UK-linked time-use rows. Because those rows conflict with the published design, the bundle excludes them pending provenance resolution. Exclusion is the correct fail-closed treatment; the anomaly does not create a UK diary target.

The FRS illustrates a different boundary. Its rounded shares describe gross weekly household income in private households, not an individual's monthly disposable resources, gaming budget, or microtransaction spending. A source can be authoritative within its own estimand while remaining unusable for another. No household-to-person bridge is supplied here.

3. Calibration, Content Addressing, and Point-Zero Design

3.1 ONS reference population

The ONS mid-2024 workbook identifies 41,860,587 UK usual residents aged 18-64, aggregated from single-year ages into five bands and divided into the source categories `FEMALE` and `MALE` (Office for National Statistics, 2025). The totals are 21,279,181 female and 20,581,406 male records across ten cells. These are population-estimate cells, not observed individual records. They do not identify gaming status, payment behaviour, harm, or correlations with income.

The runtime binding adds an optional source-recorded sex state to synthetic UK records aged 18-64. It does not infer gender identity, does not assign a sex value outside the source scope, and does not reconstruct a UK microdata population. The terminology `FEMALE` and `MALE` is retained because those are the source categories encoded by the implementation; no claim is made that they exhaustively represent sex or gender.

3.2 Coupled Hamilton allocation and SHA-256 ranking

For an age band b with n[b] in-scope synthetic records and ONS counts N[b,F] and N[b,M], the binary Hamilton quota for the female category is

$$q[b,F] = n[b] \times N[b,F] \,/\, (N[b,F] + N[b,M]),$$

with the corresponding male quota defined analogously. Floors are allocated first; any remaining unit is assigned by the larger remainder, with `FEMALE` as the documented canonical tie-break for an exact binary tie. The implementation then couples remainders across existing runtime cells within each age band so that the rounded age-band total is conserved while within-cell rounding error is minimized.

Within a runtime cell, synthetic player positions are ordered by a domain-separated SHA-256 digest over the bundle digest, age band, runtime cell identifier, and player identifier. The first positions receive the Hamilton-allocated female count and the remainder receive the male count. This deterministic rank is not random sampling from observed persons. It is a reproducible synthetic allocation of aggregate proportions. SHA-256 is also used to attest the typed population rows, the resulting sex vector, derivation inputs, assignments, runtime projection, and execution lineage (National Institute of Standards and Technology, 2015).

The audit re-attests the strict-loader output and canonically recomputes the binding. It verifies the algorithm identifier `sha256-ranked-coupled-hamilton-within-age-band-v1`, source identifier, source scope, bundle digest, typed population-weights digest, derivation-input digest, and allocation vector. A mismatch fails closed.

3.3 Lineage is not demographic fit

The binding conditions source-recorded sex within the ages already produced by the projected-population initializer. It does not resample the age margin to match ONS. Consequently, lineage attestation and demographic fit are distinct. A perfect re-attestation can coexist with a failed age-margin gate. In the present checkpoint, the same age mismatch propagates into the joint age-by-sex comparison because the conditional sex shares are applied inside the existing age distribution.

3.4 Content addressing and path portability

The bundle loader hashes declared companion files and the ignored local source cache in bounded chunks, checks byte lengths, and verifies cross-file contracts. The current audit identifies the bundle by SHA-256 digest `9b3afed444f8598f66da8ef2067d01c9921887621a9123acdc4ecbb82cefee4b`.

Hash equality establishes that the same declared bytes were used; it does not establish publisher authenticity, representativeness, construct validity, or permission to redistribute those bytes.
Runtime lineage serializes canonical JSON with sorted keys and normalized separators. Declared local path fields are normalized so that Windows backslashes and Linux forward slashes resolve to a common `<repository>/...` representation. Nested source- and population-evidence `bundle_path` values are normalized as well. Paths outside declared repository directories become `<external>`. This correction prevents identical inputs from acquiring different fingerprints solely because their checkout roots or operating systems differ. Historical fingerprints remain readable only through explicitly registered, directional legacy-to-canonical digest pairs; arbitrary digest aliasing is not allowed.

3.5 Point-zero execution scope

The checkpoint uses the exploratory configuration's seed 101 and `player_count=50000`. It initializes 50,000 synthetic player records with corresponding projected-population assignments and selects 10,024 synthetic records whose projected jurisdiction is UK and whose initialized age is 18-64 for the audit. It initializes the population projection and player-life state only. It does not initialize a scenario, execute a policy day, dispatch a campaign, compare regulatory alternatives, or generate outcome forecasts. The term *point zero* therefore denotes the initialization boundary rather than the first simulated period.

4. Audit Protocol

4.1 Definition and gate types

A point-zero audit is an audit of initialization, evidence binding, and immediate structural state before any outcome-generating transition. It is not an outcome validation exercise. The protocol uses three gate types:

1. **Structural gates** test identity, non-emptiness, binding, required fields, and zero restrictions implied by declared labels.
2. **Plausibility gates** test boundedness or distance from a declared demographic margin using engineering thresholds. They are not inferential confidence tests.
3. **Identification gates** test whether a required empirical quantity exists in a compatible typed target and is consumed by the relevant runtime mechanism.

For demographic distributions p and q over common cells c, total-variation distance is defined as

$$\mathrm{TVD}(p, q) = \tfrac{1}{2} \sum[c]\ |p[c] - q[c]|.$$

The default threshold 0.020000 is the repository-declared default engineering plausibility tolerance. It is not a sampling-error bound, confidence interval, equivalence margin justified by a substantive loss function, or proof of representativeness.

4.2 The twelve gates

Table 2. Point-zero audit gates and observed status

Gate	Type	Observed result	Status	Interpretation
Bundle attestation	Structural	Strict-loader bytes re-opened; attestation `VERIFIED`; bundle digest matched	PASS	Declared bundle, companions, and source bytes are internally consistent
Seed and cohort	Structural	Seed 101; 50,000 configured and 50,000 initialized	PASS	The requested deterministic initialization was realized

Gate	Type	Observed result	Status	Interpretation
UK selection non-empty	Structural	10,024 synthetic records with projected UK jurisdiction and initialized age 18-64	PASS	The audit population exists; no representativeness claim follows
Runtime binding	Structural	`VERIFIED_MATCH`; canonical recomputation passed	PASS	The ONS sex sidecar is bound to the declared bundle and typed rows
Runtime sex state	Structural	Explicit `FEMALE`/`MALE` values for every in-scope record; ten source cells attested	PASS	Required source-recorded state and lineage are present
Age × sex fit	Plausibility	TVD=0.033053>0.020000	FAIL	The initialized weighted joint margin exceeds the engineering tolerance
Age-margin fit	Plausibility	TVD=0.033053; maximum absolute band difference 0.021258	FAIL	The binding does not repair the initializer's age distribution
Gamer participation identified	Identification	`UNIDENTIFIED_IN_SCHEMA_V1`	FAIL	No representative typed P(gamer | age, sex) is consumed by a participation hurdle
Non-gamer play intention is zero	Structural	4,645 of 4,657 sidecar non-gamers have positive intention; serialized share `0.997423`	FAIL	The gamer/non-gamer label does not bind play intention
Non-gamer spending limit is zero	Structural	Positive share `1.000000` among 4,657 sidecar non-gamers	FAIL	The label does not bind the spending-limit initializer
Non-gamer purchase probability is zero	Structural	No explicit valid field is available and all-zero cannot be established	FAIL	No identified purchase-participation hurdle exists
Initializer finite and bounded	Plausibility	All 50,000 checked; minute fields in [0,1440] and unit fields in [0,1]	PASS	Declared numerical domains are respected

A `PASS` has only the meaning stated in its criterion. In particular, bundle attestation does not validate the data-generating process, a non-empty cohort does not imply population coverage, bounded fields do not imply behavioural realism, and a verified binding does not authorize causal, population, regulatory, or clinical inference.

5. Results: The Negative Result

The primary result is the correct blocking of empirically labelled execution, not a software failure. The audit reports `6 passed / 6 failed / 12 total` and the overall state `FAIL_CLOSED`. The command returns exit code 1 by design whenever a gate fails or an error occurs. Its execution-scope record confirms that no scenario, policy day, or campaign was executed.

Table 3. Principal point-zero quantities and epistemic status

Quantity	Value	Source or derivation	Epistemic status
UK residents aged 18-64	41,860,587	ONS mid-2024 cells	Official population margin used for calibration infrastructure
Initialized synthetic cohort	50,000	Seed-101 configuration and initializer	Deterministic computational state; not a UK sample

Quantity	Value	Source or derivation	Epistemic status
Selected synthetic UK-labelled records aged 18-64	10,024	Point-zero filter	Deterministic audit subset
Age × sex TVD	0.033053	Initialized weighted cells versus ONS	Failed engineering plausibility diagnostic
Maximum absolute age-band difference	0.021258	Maximum of absolute share differences	Failed demographic diagnostic; largest band is 55-64
P(gamer | age, sex)	Not identified	No compatible representative target/hurdle	Non-identification result
Positive play intention among sidecar non-gamers	4,645/4,657 = 0.9974232338	Initialized arrays	Failed structural binding diagnostic
Positive spending limit among sidecar non-gamers	100%	Initialized arrays	Failed structural binding diagnostic
Explicit zero purchase probability for non-gamers	Absent	Runtime schema/state	Failed structural binding diagnostic
`simulation_cents` to GBP	Not identified	Bundle record	Unquantified; real-money inference disabled
Composite-harm weights and high-risk threshold	Not externally identified	Model configuration	Normative or unsupported; no clinical inference

5.1 Demographic discrepancy

The weighted age-by-sex TVD is 0.033053, exceeding 0.020000. The age-margin TVD is the same. Band share differences, initialized minus ONS, are +0.011038 for ages 18-24, -0.011795 for 25-34, +0.001617 for 35-44, +0.020398 for 45-54, and -0.021258 for 55-64 after rounding to six decimals. The largest absolute difference is therefore 0.021258. The repository attributes this pattern to the existing broader 55-69 source band and uniform within-band age sampling. Because the point-zero bridge only conditions sex within the realized age bands, it cannot correct this discrepancy.

The runtime sex vector contains 5,093 `FEMALE` and 4,931 `MALE` assignments among the 10,024 in-scope records; 39,976 out-of-scope records remain unavailable. Coupled Hamilton allocation preserves the Hamilton-rounded female allocation implied by the ONS female/male proportions within each realized runtime age band; male assignments are the complement. That arithmetic success is a property of the binding, not evidence that the age distribution or individual assignments are observed.

5.2 Gamer participation and behavioural binding

The bundle does not identify `P(gamer | ONS age-by-sex cell)`, and no typed participation target is consumed by a runtime hurdle. The projected population contains illustrative `baseline_gamer` sidecar metadata: 5,367 selected records are labelled gamer and 4,657 non-gamer. The audit does not treat those labels as empirical UK participation estimates.

The sidecar labels are not behaviourally binding. Of the 4,657 labelled non-gamers, 4,645 receive positive `intended_play_minutes`, giving an exact share of 0.9974232338, or 99.7423%. The checked-in narrative rounds this figure as 99.743%; direct conversion of the underlying count gives 99.742% to three decimal places. This manuscript retains the canonical serialized share `0.997423`, the underlying count, and the documentation discrepancy rather than silently selecting one representation. All 4,657 sidecar non-gamers receive a positive initialized spending limit. No explicit, valid `purchase_probability` field exists that can be verified as zero for every non-gamer.

These failures do not imply that every non-gamer in the real world must have zero latent interest or that all gaming classifications are immutable. They test a narrower implementation contract: if a runtime label is invoked as a participation state, it must be linked to the relevant behaviour or explicitly remain non-binding metadata. The current labels do the latter, so they cannot support behavioural calibration claims.

5.3 Why the time-use figures are not calibration residuals

The time-use comparisons are diagnostic discrepancies between incommensurable constructs. They are not residuals from a common measurement model, and their ratios are not calibration coefficients.

Table 4. Measurement dictionary for the principal play-time quantities

Quantity	Value	Population/denominator	Construct	Permitted use
ONS gaming	16.8 minutes/day	UK adults 18+, gamers and non-gamers	Realised unconditional primary-activity diary time	Diagnostic benchmark only
Ofcom/Ampere play	417 minutes/week	UK gamers aged 16-64, Q2 2024	Gamer-reported weekly online or offline play; no figure-level uncertainty published	External diagnostic/declared validation source, not a passed validation result
Open Play intake play	1,004.269880 minutes/week	Selected qualified UK participants aged 18-40, valid response n=830	Self-reported weekly play	Non-representative conditional diagnostic
Simulator all selected	82.828811 minutes/day	Synthetic UK 18-64 audit subset	Initialized play intention	Computational state, not realised play
Simulator sidecar gamers	581.476057 minutes/week	Synthetic UK 18-64 records labelled gamer	Initialized play intention	Structural diagnostic only

First, the all-selected simulator-to-ONS ratio is

$$82.828811 \text{ minutes/day of initialized play intention} \div 16.8 \text{ minutes/day of realised unconditional ONS gaming} = 4.930286.$$

Second, the sidecar-gamer simulator-to-Ofcom/Ampere ratio is

$$581.476057 \text{ minutes/week of initialized play intention} \div 417 \text{ minutes/week of gamer self-report} = 1.394427.$$

Third, the Open Play-to-Ofcom/Ampere source comparison is

$$1{,}004.269880 \text{ minutes/week in selected Open Play } (\mathrm{n} = 830) \div 417 \text{ minutes/week in Ofcom/Ampere} = 2.408321.$$

The third ratio is a source-to-source diagnostic discrepancy, not an emitted point-zero gate and not a simulator residual. The compact target reports n = 830 after a manuscript-aligned filter excluding 19 boundary responses at exactly 3,000 minutes/week; that local extraction is not reconstructible from checked-in code alone. Open Play uses a selected trace-sharing cohort, not a representative UK sample, and its value cannot be transported as a UK population target. Ofcom/Ampere uses a different age range, a gamer-only denominator, and self-report with no published uncertainty for the cited mean. ONS uses an unconditional adult diary measure. The appropriate language is therefore *incommensurability*, *non-transportability*, and *diagnostic discrepancy*, not overestimation or bias relative to a shared gold standard.

6. Fail-Closed as a Scientific Design Principle

A fail-closed design prevents illustrative assumptions from becoming apparent evidence through procedural momentum. In the present repository, the audit can attest exact bytes, identities, and structural state and still refuse campaign authorization. This separation is scientifically useful because a deterministic pipeline can otherwise make an unsupported result appear more credible than a visibly incomplete one.

Four activities should remain distinct:

1. **Auditing and attestation** establish what files, configuration, algorithms, and runtime state were used and whether their declared identities match.
2. **Calibration** estimates or assigns parameters against semantically compatible targets under an explicit loss function and uncertainty model.
3. **Out-of-sample validation** evaluates fixed predictions or measurements against genuinely held-out units or sources using predeclared thresholds.
4. **Campaign authorization** permits a bounded simulation campaign only after independent data, model, monetary, causal, and interpretive gates are satisfied.

The current work completes part of the first activity and a limited demographic plausibility check. It does not complete behavioural calibration, out-of-sample outcome validation, or campaign authorization. A `VALIDATION` label in a target table records intended evidentiary role; it is not evidence that validation occurred.

Stopping before treatment execution can itself be a methodological result when the stopping criteria, code, and diagnostics are explicit. The result identifies the prerequisite failures encoded by this audit and makes the no-go decision reproducible under the local data conditions. It does not prove that the model is correct, that all untested components are defective, or that a different data architecture could not pass. Fail-closed operation is a constraint against unsupported inference, not a validation certificate.

7. Limitations

1. **Aggregate and misaligned sources do not identify individual behaviour.** ONS population cells, FRS household-income margins, Ofcom/Ampere gamer self-reports, and selected Open Play records do not jointly observe the same persons or constructs. Their combination would require explicit data-fusion and transport assumptions.
2. **A demographic margin is not behavioural calibration.** Matching or binding age and source-recorded sex does not calibrate gaming participation, purchase participation, spending, transaction tails, harm, or policy response. The present age margin itself fails the engineering tolerance.
3. **Open Play is selected, and its UK-linked time-use rows are excluded.** The study requires willingness to share traces and complete an intensive protocol, covers UK ages 18-40, and is not representative of the general population or all adults who play games. The 43 UK-linked time-use rows conflict with the published US-only diary design and remain outside calibration and validation.
4. **Ofcom/Ampere is gamer-only and differently scoped.** The 417-minute value is self-reported weekly play among gamers aged 16-64, not an unconditional UK adult measure. The report does not publish uncertainty for the cited figure. Its role label does not create a compatible validation operator.
5. `simulation_cents` **has no empirical bridge to GBP.** Official FX, inflation, and purchasing-power series cannot identify the conversion from an internal model unit to observed microtransaction spending. Real-money, national-expenditure, and monetized-harm claims are therefore unauthorized.
6. **Harm weights and the high-risk threshold remain normative or unidentified.** The composite weights are value choices, and the hard-coded high-risk rule has no representative UK adult

prevalence or external clinical validation in the bundle. Screening scores cannot be relabelled as diagnoses, and this audit makes no clinical claim.

7. **Public rerun is incomplete without the excluded source cache.** The repository versions code, documentation, compact derivatives, bundle declarations, and digests, but the complete local source cache required for strict re-attestation is intentionally ignored because it contains participant-level material and licence-constrained files. In addition, no checked-in extractor rebuilds all 75 target rows. Public computational reproducibility must therefore be described as partial and conditional, not full end-to-end reproducibility.

8. Requirements for Future Research

This section is a roadmap, not a report of completed studies or anticipated empirical results. Each element would require a successor bundle, analysis plan, and independent authorization before fitting.

8.1 Data donation

A data-donation study should begin with ethics approval, informed consent, a documented lawful basis, data minimization, secure transfer and storage, retention and deletion rules, withdrawal procedures, and disclosure-risk review. The study should not assume that a subject-access export provides a standardized research table. Right-of-access guidance concerns access to personal information; it does not require a standardized research export containing every field relevant to a study (European Data Protection Board, 2023; Information Commissioner's Office, 2026).

Small platform-specific pilots should inventory the fields actually returned and the periods covered. The pilot should test whether exports expose transaction timestamps, gross and net amounts, currencies, refunds, regional prices, SKU or item identifiers, bundles, subscriptions, virtual-currency conversions, and offer exposures. It should distinguish purchases from catalogue availability and should not infer rejected alternatives from items that happened to be listed.

Consent and upload friction must be measured. Prior data-donation research shows that willingness does not imply successful donation and that nonparticipation can be selective (Keusch et al., 2024). A future study should report invitation, consent, download, upload, parse, and usable-record rates; compare donors and non-donors where possible; and examine coverage of zero-spenders. These quantities are design outcomes, not nuisances to be assumed away.

8.2 Ecological momentary assessment

A 14-30 day ecological momentary assessment (EMA) could measure spending intention close to the session rather than reconstructing it from a budget rule. Before each session, the study should record whether the participant intends to spend and the maximum planned amount. Immediately after the session, it should record actual spending, context, offer type, and reported trigger. With separate consent, transaction records could be reconciled to the post-session report by timestamp and amount.

EMA reduces reliance on global retrospective summaries but does not eliminate measurement error or participation effects (Shiffman, Stone, & Hufford, 2008). The analysis plan should specify missingness, compliance, reactivity, burden, and attrition analyses; rules for overlapping or unclosed sessions; treatment of zero-spend sessions; and reconciliation tolerances. The planned/unplanned distinction should be measured, not defined algebraically from the same spending outcome it is intended to explain.

8.3 Same-person UK measurement

A future UK sample should measure play, microtransactions, sleep, interference with study or work, financial strain, and impairment in the same individuals. It should preserve the conditioning sets for non-gamers, gamers who do not spend, occasional spenders, subscribers, refunded purchases, and highfrequency purchasers. Sampling should cover the intended 18-64 resident universe or explicitly redefine the target population; age, sex/gender, and participation measures should not be imported from separate surveys as if jointly observed.

Clinical or screening measures must retain their validated interpretation. A screening score may describe symptom severity under its instrument but does not establish diagnosis, causality, or a uniquely correct simulator threshold. Cross-sectional same-person measurement would improve construct linkage while still requiring a causal design for causal claims.

8.4 Identification, measurement operators, and holdouts

Future calibration requires an empirical target for

$$P(\text{gamer} \mid \text{age}, \text{sex}),$$

defined for the same age bands, population universe, and source categories used by the runtime. The target must enter a typed participation hurdle rather than remain sidecar metadata. Purchase participation and conditional spending would require their own denominators, periods, zero-spender treatment, and uncertainty.

Every fitted source needs a semantically compatible measurement operator. If the model produces intention, validation must either observe intention or specify and estimate a bridge from intention to realised behaviour. Session-like simulator steps cannot be compared directly with inferred platform polling segments without an observation model. Aggregate weekly self-report, diary minutes, and telemetry records should remain separate unless an explicitly validated transformation connects them.

Validation and holdout partitions must contain genuinely independent source units or prospectively fixed row sets, not merely different free-text locators. Canonical selectors or row-set digests should bind the partitions. A preregistration should freeze estimands, source and target populations, transformations, loss functions, uncertainty, thresholds, seeds, exclusion rules, missing-data handling, and stop criteria before any fit or inspection of holdout performance.

Data and Code Availability

The audited repository is available at https://github.com/francescoromano1407-stack/microtx-sim. The point-zero results reported here were re-executed on 1 September 2026 using the tracked code at commit `58501bc0fb799f43970f2f9fc99031c555ab06f2`. The repository contains source code, tests, documentation, the `uk-adults-2024-v1` compact bundle, companion files, source metadata, and SHA-256 digests. It also contains `uv.lock`; the project declares Python >=3.11 and pins the `uv` tool version to 0.11.6. The local rerun used Windows 11 build 26200, Python 3.12.13, NumPy 2.5.2, and uv 0.11.6.

This is not a claim of full public end-to-end reproducibility. The complete cache at `data/public_calibration_sources_uk_adults_2024/` is intentionally excluded from version control because it contains third-party source archives, including publicly released pseudonymized Open Play data subject to its no-reidentification and responsible-use conditions, and other source files governed by their respective reuse terms. The strict point-zero command requires that local hash-attested cache. No checked-in extractor currently reconstructs all 75 target rows from the original workbooks and participant files, and the command-emitted point-zero JSON is not itself versioned in the audited commit. The command python tools/run_point_zero_audit.py was rerun locally on 1 September 2026. Its canonical compact UTF-8 JSON comprised 10,531 bytes and had SHA-256 1b5f4600af1709fdd8349d9fa646fe327abc4a68ff5916b00930c4eadc74df03.

Public users without the excluded cache can inspect the code, compact derivatives, tests, and declarations, but should not be promised an identical strict-cache rerun from the public repository alone.

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Conflicts of Interest

The author declares no conflicts of interest.

Ethics Statement

This study is a secondary computational audit of pre-existing, publicly available materials. Open Play v1.2.5 was accessed from its version-specific Zenodo release (https://doi.org/10.5281/zenodo.21986572). The archived release declares open access and CC0 1.0 Universal in its zenodo.json file; its bundled LICENSE retains CC0 1.0 Universal and adds no-reidentification and responsible-use conditions. The source repository describes the raw files as pseudonymized.

No new data were collected, no participants were contacted, and no attempt was made to re-identify, deanonymize, or link records to individuals. This manuscript and the accompanying microtx-sim repository do not redistribute the Open Play archive, participant-level records, or row-level derivatives. They report only aggregate diagnostics, compact derivatives, source metadata, code, and cryptographic digests. The local cache was used solely for source-byte re-attestation and remains outside version control.

This audit does not assess the original study's consent, legal basis, or ethics-review arrangements. It makes no clinical, diagnostic, individual-level, causal, or population-prevalence claims.

Author Contributions

Francesco Romano: Conceptualization; Methodology; Software; Validation (computational verification only); Formal Analysis; Investigation; Data Curation; Writing – Original Draft; Writing – Review & Editing; Visualization; Project Administration. The author is solely responsible for all aspects of the work.

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Appendices

Appendix A. Gate definitions and decision rules

Table A1. Formal point-zero gate definitions

Gate identifier	Type	Decision rule	What failure means
`calibration_bundle_attested`	Structural	Re-open the strict-loader output; require exact bundle, companion, and source-byte attestation	Declared evidence identity cannot be trusted for this run
`configured_seed_and_cohort`	Structural	Require seed 101 and initialized count equal to configured 50,000	The declared checkpoint was not reproduced
`uk_adult_selection_nonempty`	Structural	Require at least one synthetic record with projected UK jurisdiction and initialized age 18-64	No in-scope audit population exists
`calibration_bundle_runtime_binding`	Structural	Require typed bundle ID/digest, source ID, weights digest, assignment method, jurisdiction and age scope, and canonical recomputation to match	The runtime sidecar is not the declared ONS binding
`runtime_sex_state`	Structural	Require every in-scope record to have `FEMALE` or `MALE` and the vector to enter execution lineage	Required source-recorded state or lineage is absent
`ons_age_sex_joint_fit`	Plausibility	Require weighted ten-cell TVD ≤ 0.020000	Joint demographic discrepancy exceeds the engineering tolerance
`ons_age_marginal_fit`	Plausibility	Require weighted five-band TVD ≤ 0.020000	Age discrepancy exceeds the engineering tolerance
`gamer_participation_identified`	Identification	Require a representative typed P(gamer | age, sex) target consumed by a participation hurdle	Gamer status is not empirically identified in runtime behaviour
`non_gamer_zero_play_intention`	Structural	Require zero `intended_play_minutes` for every sidecar non-gamer	Sidecar gamer state does not bind intended play
`non_gamer_zero_spending_limit`	Structural	Require zero `intended_spending_limit_cents` for every sidecar non-gamer	Sidecar gamer state does not bind the spending-limit initializer
`non_gamer_zero_purchase_probability`	Structural	Require an explicit valid field equal to zero for every sidecar non-gamer	No verifiable purchase-participation hurdle exists
`initializer_state_finite_and_bounded`	Plausibility	Require minute fields finite in [0,1440] and unit fields finite in [0,1] for all 50,000 records	Initialization violates declared numerical domains

The audit state is `PASS` only if all 12 gates pass; otherwise it is `FAIL_CLOSED`. Errors also return a fail-closed status and exit code 1. Gate aggregation is conjunctive; one failed gate is sufficient to stop empirically labelled execution.

Appendix B. ONS age × sex cells, Hamilton allocation, and TVD

Table B1. ONS mid-2024 UK usual residents aged 18-64

Age band	Female	Male	Persons	Person share
18-24	2,821,237	2,970,284	5,791,521	0.138352599
25-34	4,754,911	4,590,686	9,345,597	0.223255278
35-44	4,805,400	4,499,492	9,304,892	0.222282884

Age band	Female	Male	Persons	Person share
45-54	4,343,516	4,169,961	8,513,477	0.203376914
55-64	4,554,117	4,350,983	8,905,100	0.212732325
Total 18-64	**21,279,181**	**20,581,406**	**41,860,587**	**1.000000000**

For each age band b, let n[bk] be the count in runtime cell k and let r[b]=N[b,F]/(N[b,F]+N[b,M]). The implementation computes exact rational female quotas q[bk]=n[bk]r[b], allocates ⌊ q[bk]⌋, and assigns the remaining age-band female units to cells with the largest fractional remainders. Domain-separated SHA-256 cell ranks break remainder ties. Within each cell, another domain-separated SHA-256 rank determines which synthetic player positions receive the allocated category. Conservation requires

$$\sum[k]\ h[b,k] = H[b],$$

where h[bk] is the allocated female count in cell k and H[b] is the binary Hamilton female count for the complete age band.
For the joint comparison, cell proportions are normalized over the ten common age-by-sex cells:

$$\mathrm{TVD}[\mathrm{age} \times \mathrm{sex}] = ½ \sum[b \in B] \sum[s \in \{F, M\}]\ |p_{ru}^{ntime}[b,s] - p^{ons}[b,s]|.$$

For the age margin,

$$\mathrm{TVD}[\mathrm{age}] = ½ \sum[b \in B]\ |p_{ru}^{ntime}[b] - p^{ons}[b]|.$$

Both equal 0.033053 in the audited checkpoint because the sex binding applies ONS conditional proportions inside the already realized runtime age bands and does not resample age.

Appendix C. Measure dictionary and comparability matrix

Table C1. Comparability across ONS, Ofcom/Ampere, Open Play, and simulator state

Dimension	ONS March 2024	Ofcom/Ampere Q2 2024	Open Play v1.2.5	Simulator point zero
Population	UK adults 18+	UK gamers 16-64	Selected qualified UK participants 18-40	Synthetic selected UK records 18-64
Conditioning	Unconditional; gamers and non-gamers	Gamer-only	Study participation, qualification, UK, age, valid intake response	All selected or illustrative sidecar-gamer subset
Measure	Realised primary-activity gaming	Self-reported weekly play online/offline	Self-reported weekly play; platform traces are separate measures	Initialized intended play
Unit	Minutes/day	Minutes/week	Minutes/week	Minutes/day, converted descriptively to week
Central value used	16.8	417	1,004.269880, n=830	82.828811 all; 581.476057/week sidecar gamers
Sampling/ coverage	Official statistics in development; all-adult diary estimate	Ampere survey figure reported by Ofcom	Selected trace-sharing cohort; not representative	Deterministic synthetic initialization
Figure uncertainty	Published interval 13.5-20.0	Not published for cited mean	Not used as a population interval	No sampling interpretation
Runtime role	Not connected	Not connected	Excluded or not connected	Audited state

Dimension	ONS March 2024	Ofcom/Ampere Q2 2024	Open Play v1.2.5	Simulator point zero
Comparable as calibration residual?	No	No	No	Not applicable

The 43 UK-linked Open Play time-use rows are not included in this matrix as a UK diary measure. They are excluded because their provenance conflicts with the published US-only diary design. Nintendo and Steam record durations are also not session targets: the Steam rows are inferred polling segments from cumulative deltas, Nintendo covers a restricted platform set, and repeated rows are not independent observations.

Appendix D. Reproducibility, digests, path normalization, and cache limits

Table D1. Selected identities from the audited run

Object	SHA-256 or identity	Function
Git commit	`58501bc0fb799f43970f2f9fc99031c555ab06f2`	Audited code state
Calibration bundle	`9b3afed444f8598f66da8ef2067d01c9921887621a9123acdc4ecbb82cefee4b`	Strict bundle identity
`targets.csv` declared bytes	`0747d049e8fcb392a8306a856baebebfc1e240f17371c50486740f1aa9b23c7f`	Companion-file attestation
`population_weights.csv` declared bytes	`127627d70f56e2f056b449aa317474c103946a95c997e252f170e1553e79be23`	Companion-file attestation
Typed population-weights payload	`7a82d6a3e4281aa9ecbaf8904ec33e10fe998c50da393f443fae3292550b608f`	Exact runtime-consumed row identity
`source_manifest.json` declared bytes	`a732dec2093e3ab97c7f7f3ee50c6ad454fd36c5156f3e59c4f82ecb06d8a63d`	Companion-file attestation
Source-recorded assignment	`a2dcea22acf10d6a44287de56e1e6fbee4e123b8d63381b110fa8cd60d591d13`	Cohort assignment identity
Runtime projection	`0031a65cc6b5bdde3445324cfc107ee7872be2247ab3782637cae61989da3267`	Projection state identity
Population execution	`d8fd98ad1291586da40937757a819892f39510891184f74fa6acf3a4faf0a38e`	Execution-lineage identity

The source manifest records verified_at=2026-09-01, which attests local byte/hash verification rather than a public rerun. The point-zero command is designed to return exit code 1 with FAIL_CLOSED when any gate fails and to return ERROR_FAIL_CLOSED with exit code 1 on errors. The current tests/test_point_zero_audit.py module contains 12 test cases. These software checks establish declared computational properties; they are not empirical outcome validation. The validator's blocker text is partly stale: it still states that no dedicated point-zero command exists and that age-sex targets are not consumed, although the repository contains the audited command and point-zero sex binding. This documentation inconsistency should be corrected before deposit; it does not change campaign_ready=false or the six-gate FAIL_CLOSED result.

The profile-input lineage fingerprint uses canonical JSON with sorted keys, compact separators, finite-number enforcement, and UTF-8 encoding. Portability normalization replaces both Windows and POSIX separators, identifies the last declared repository-directory component, and emits `<repository>/...`; paths not mapped to the repository become `<external>`. The normalization is limited to declared path fields, including nested evidence `bundle_path` fields. It does not alter source bytes or accept arbitrary digest substitutions.

The ignored local cache is necessary for strict source re-attestation. It contains original workbooks, PDFs, archives, and participant-level Open Play files. Publishing hashes does not confer redistribution permission, remove privacy risk, or allow a third party to reconstruct missing bytes. The archived Open Play release declares open access and CC0 1.0 Universal in zenodo.json. Its bundled LICENSE retains CC0 1.0 Universal and adds no-reidentification and responsible-use conditions; this audit follows those additional conditions. A public deposit should therefore include only permissible compact derivatives and code, plus a precise access or reconstruction statement; it should not claim that the public repository alone reproduces the strict-cache audit.